

WEEK 8

STUDENT PERFORMANCE DASHBOARD

Introduction :

The Student Performance Dashboard was created using Power BI to analyze students' academic performance.

The dashboard helps to understand students' performance in Math, Reading, and Writing and compare performance based on demographic and academic factors such as Gender, Race/Ethnicity, Parental Level of Education, Lunch Type, and Test Preparation Course.

Interactive slicers, charts, KPI cards, and a drill-through page were used to make the dashboard easy to analyze.

Objectives :

The main objectives of this dashboard are:

- To analyze the overall academic performance of students.
- To compare Math, Reading, and Writing scores.
- To identify differences in performance across Race/Ethnicity.
- To analyze performance based on Parental Level of Education.
- To compare students based on Test Preparation Course.
- To analyze performance based on Lunch Type.
- To understand Grade distribution.
- To provide interactive filtering using slicers.
- To provide detailed student-level information using Drill-through.

Dashboard Development :

The dashboard was developed using Power BI.

The dashboard consists of four pages:

1. Page 1 – Overall Performance
2. Page 2 – Performance Analysis
3. Page 3 – Detailed Analysis
4. Page 4 – Academic Profile (Drill-through)

Page 1 – Overall Performance :

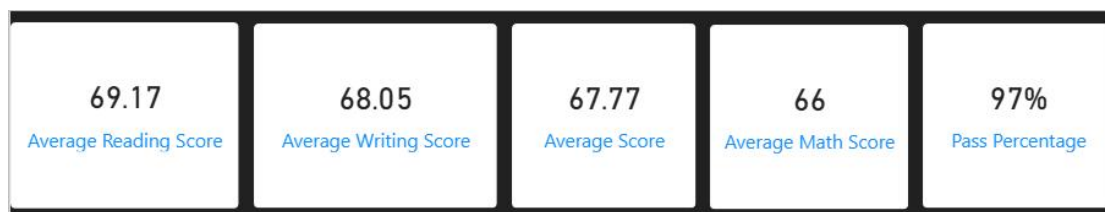
Page 1 provides an overall view of student performance using KPI cards and comparison charts.

➤ KPI Cards :

Five KPI cards were created:

- Average Reading Score
- Average Writing Score
- Average Score
- Average Math Score
- Pass Percentage

These cards provide a quick summary of the overall academic performance.



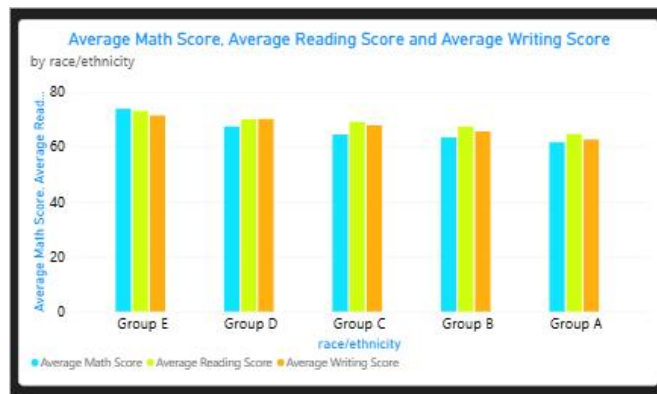
➤ Race/Ethnicity Performance :

A Clustered Column Chart was created to compare subject-wise performance across Race/Ethnicity.

Fields used:

- **X-axis:** Race/Ethnicity
- **Y-axis:** Average Math Score
- **Y-axis:** Average Reading Score
- **Y-axis:** Average Writing Score

This visual helps compare the three subject averages across different race/ethnicity groups.



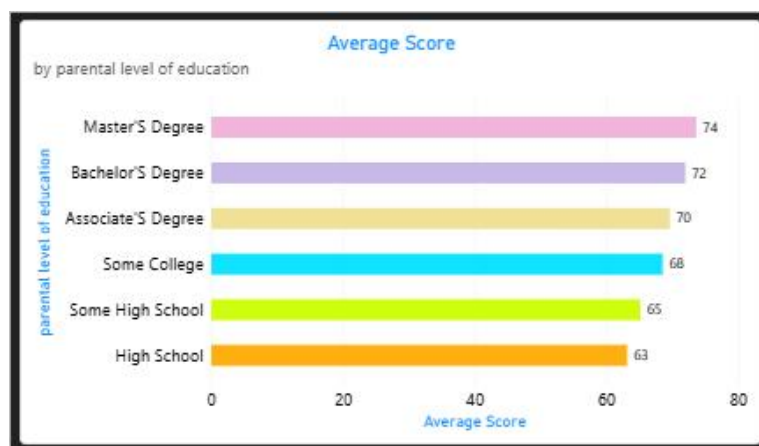
➤ Parental Level of Education :

A Clustered Bar Chart was created to analyze average student scores based on parental education level.

Fields used:

- **Y-axis:** Parental Level of Education
- **X-axis:** Average Score

This visual helps identify how average student performance differs across parental education levels.



Page 2 – Performance Analysis :

Page 2 focuses on test preparation, lunch type, grade distribution, and interactive filtering.

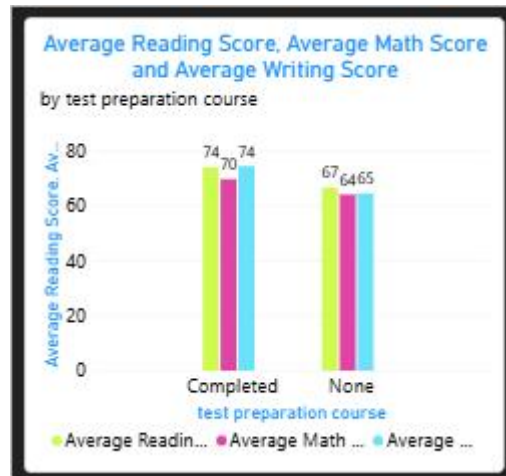
➤ Test Preparation Course :

A Clustered Column Chart was created to compare subject performance based on the Test Preparation Course.

Fields used:

- **X-axis:** Test Preparation Course
- **Y-axis:** Average Math Score
- **Y-axis:** Average Reading Score
- **Y-axis:** Average Writing Score

This visual helps compare the academic scores of students based on their test preparation status.



➤ Lunch Type – Average Score :

A Clustered Column Chart was used to compare average scores based on lunch type.

Fields used:

- **X-axis:** Lunch
- **Y-axis:** Average Score

This allows comparison of average academic performance between different lunch categories.



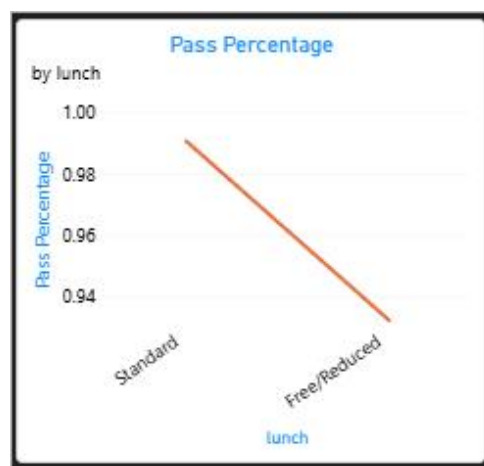
➤ Lunch Type – Pass Percentage :

A Line Chart was created to compare pass percentage across lunch categories.

Fields used:

- **X-axis:** Lunch
- **Y-axis:** Pass Percentage

This provides a comparison of pass percentage between the lunch groups.



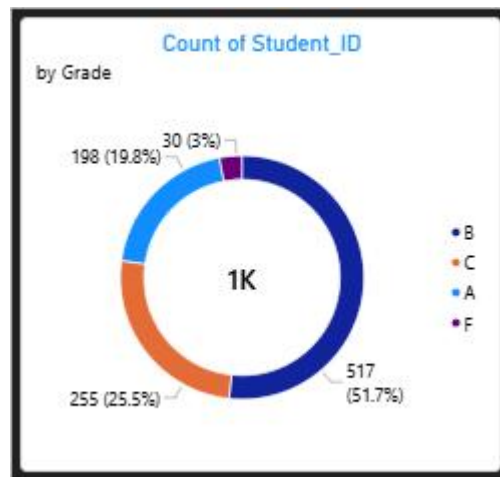
➤ Grade Distribution :

A Donut Chart was created to show the distribution of students across grades.

Fields used:

- **Legend:** Grade
- **Values:** Count of Student_ID

This visual shows how students are distributed across different grade categories.



Slicers and Interactive Filtering :

Multiple slicers were added to make the dashboard interactive.

The slicers used are:

1. Test Preparation Course
2. Lunch
3. Gender
4. Grade
5. Race/Ethnicity
6. Subject
7. Parental Level of Education
8. Student_ID

The slicers were formatted using Dropdown and Vertical List styles.

➤ Slicer Format :

✧ Dropdown slicers:

- Test Preparation Course
- Lunch
- Gender
- Grade
- Race/Ethnicity
- Subject

✧ Vertical List slicers:

- Parental Level of Education

- Student_ID

The slicers were synchronized across Page 1, Page 2, and Page 3, allowing consistent filtering across the main dashboard pages.

test preparation course All ▼	lunch All ▼
gender All ▼	Grade All ▼
race/ethnicity All ▼	Subjects All ▼
parental level of education ☐ Select all ☐ Associate'S Degree ☐ Bachelor'S Degree ☐ High School ☐ Master'S Degree ☐ Some College ☐ Some High School	Student_ID ☐ Select all ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6

Page 3 – Detailed Analysis :

Page 3 contains additional visual analysis of lunch, gender, subjects, test preparation, and individual student performance.

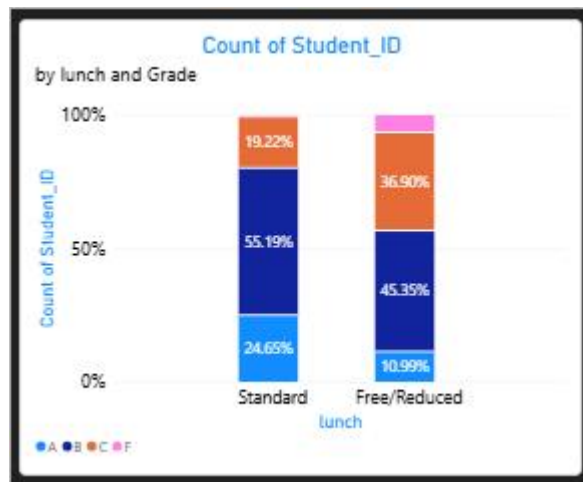
➤ Grade Distribution by Lunch :

A Stacked Column Chart was created.

Fields used:

- **X-axis:** Lunch
- **Y-axis:** Count of Student_ID
- **Legend:** Grade

This visual helps compare grade distribution across different lunch categories.



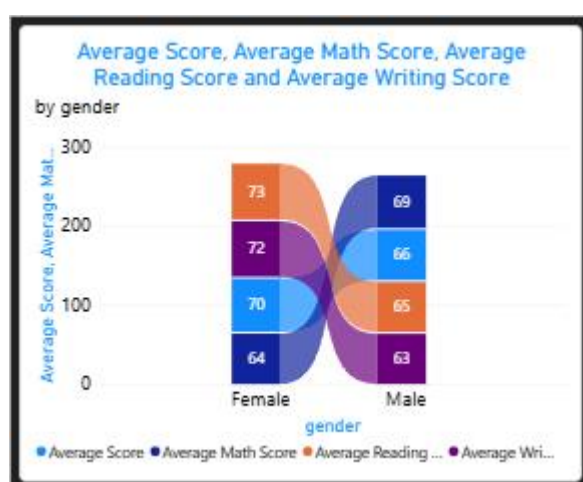
➤ **Gender-wise Performance :**

A Ribbon Chart was created to compare student performance by gender.

Fields used:

- **X-axis:** Gender
- **Y-axis:** Average Score
- **Y-axis:** Average Math Score
- **Y-axis:** Average Reading Score
- **Y-axis:** Average Writing Score

This visual provides a comparative view of different performance measures across gender categories.



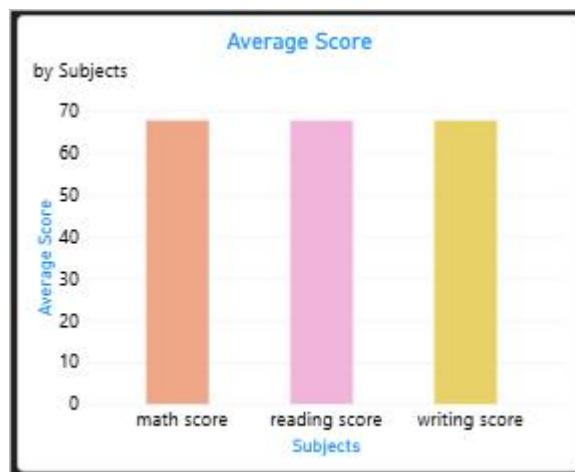
➤ **Subject Performance :**

A Clustered Column Chart was created to compare average performance across subjects.

Fields used:

- **X-axis:** Subject
- **Y-axis:** Average Score

This visual helps identify differences between subject-wise average performance.



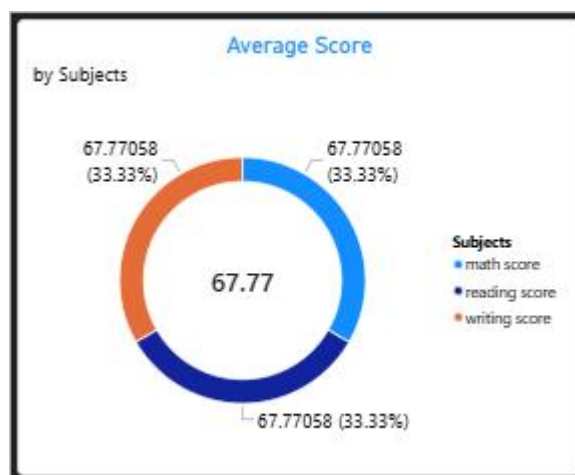
➤ Subject-wise Average Score :

A Donut Chart was created using subject and average score.

Fields used:

- **Legend:** Subject
- **Values:** Average Score

This provides another visual representation of subject-wise average performance.



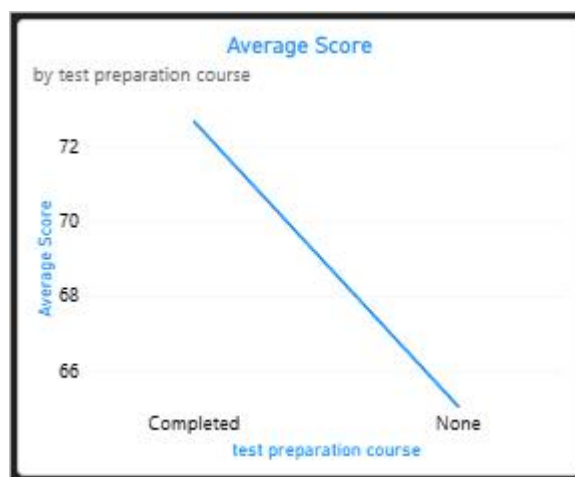
➤ Test Preparation and Average Score :

A Line Chart was created to compare average scores based on test preparation status.

Fields used:

- **X-axis:** Test Preparation Course
- **Y-axis:** Average Score

This helps compare the average performance of students with different test preparation statuses.



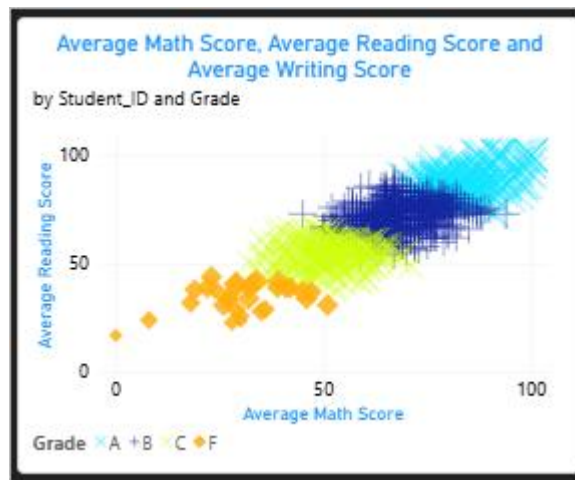
➤ Student Performance Scatter Plot :

A Scatter Chart was created to examine the relationship between Math and Reading scores.

Fields used:

- **X-axis:** Average Math Score
- **Y-axis:** Average Reading Score
- **Legend:** Gender
- **Size:** Average Writing Score
- **Values:** Student_ID

This visual allows individual student performance to be viewed based on Math and Reading scores while also representing Gender and Writing performance.



Page 4 – Academic Profile / Drill-through :

A separate Academic Profile page was created using the Power BI Drill-through feature.

Drill-through allows the user to move from a general dashboard visual to a detailed page for a selected student.

Fields included in the Academic Profile table:

- Student_ID
- Gender
- Subject
- Average Score
- Grade
- Result
- Test Preparation Course
- Parental Level of Education
- Lunch

When a student is selected from the dashboard and Drill through - Academic Profile is used, the Academic Profile page displays the details of that selected student.

A Back Arrow was also added to return to the previous page.

Before Drill Through :

DRILL THROUGH PAGE								
Student_ID	gender	Subjects	Average Score	Grade	Result	test preparation course	parental level of education	lunch
1	Female	math score	72.67	B	Pass	None	Bachelor'S Degree	Standard
1	Female	reading score	72.67	B	Pass	None	Bachelor'S Degree	Standard
1	Female	writing score	72.67	B	Pass	None	Bachelor'S Degree	Standard
2	Female	math score	82.33	A	Pass	Completed	Some College	Standard
2	Female	reading score	82.33	A	Pass	Completed	Some College	Standard
2	Female	writing score	82.33	A	Pass	Completed	Some College	Standard
3	Female	math score	92.67	A	Pass	None	Master'S Degree	Standard
3	Female	reading score	92.67	A	Pass	None	Master'S Degree	Standard
3	Female	writing score	92.67	A	Pass	None	Master'S Degree	Standard
6	Female	math score	77.33	B	Pass	None	Associate'S Degree	Standard
6	Female	reading score	77.33	B	Pass	None	Associate'S Degree	Standard
6	Female	writing score	77.33	B	Pass	None	Associate'S Degree	Standard
7	Female	math score	91.67	A	Pass	Completed	Some College	Standard
7	Female	reading score	91.67	A	Pass	Completed	Some College	Standard
7	Female	writing score	91.67	A	Pass	Completed	Some College	Standard
10	Female	math score	49.33	C	Pass	None	High School	Free/Reduced
10	Female	reading score	49.33	C	Pass	None	High School	Free/Reduced
Total			67.77					

After Drill Through :

Student_ID	gender	Subjects	Average Score	Grade	Result	test preparation course	parental level of education	lunch
3	Female	math score	92.67	A	Pass	None	Master'S Degree	Standard
3	Female	reading score	92.67	A	Pass	None	Master'S Degree	Standard
3	Female	writing score	92.67	A	Pass	None	Master'S Degree	Standard
Total			92.67					

Interactive Features :

The dashboard includes the following interactive features:

- Interactive KPI cards
- Multiple slicers
- Slicer synchronization
- Student-level Drill-through
- Back navigation button
- Interactive charts

These features allow users to explore student performance from different perspectives.

Key Insights :

- ✧ **Overall Performance:** Students show good overall academic performance.
- ✧ **Subject Performance:** Average scores vary across Math, Reading, and Writing.
- ✧ **Test Preparation:** Students who completed test preparation generally show better scores.
- ✧ **Lunch Type:** Performance differs between the lunch groups.
- ✧ **Parental Education:** Average scores vary based on parents' education level.

Conclusion :

The Student Performance Dashboard provides an interactive way to analyze student academic performance using Power BI. It combines KPI cards, charts, slicers, interactive filtering, and Drill-through functionality to provide both overall and student-level analysis.

The dashboard makes it easier to compare subject performance and explore academic performance based on different student characteristics.